

# ERDŐS PROBLEM 501 AFTER ADDING $\omega_2$ RANDOM REALS

ABSTRACT. Let  $P$  be the positive assertion in Erdős problem 501: every family  $(A_y)_{y \in \mathbb{R}}$  of bounded subsets of  $\mathbb{R}$  with  $\lambda^*(A_y) < 1$  has an infinite set  $X \subseteq \mathbb{R}$  such that  $x \notin A_y$  whenever  $x, y \in X$  are distinct. We prove that  $P$  holds in the extension of any model of  $\text{ZFC} + \text{CH}$  by  $\omega_2$  random reals. In fact, boundedness is unnecessary. Following Lee's measure-theoretic double-counting framework, we first prove a Tonelli selection lemma for measurable directed graphs. To apply it to an arbitrary family  $(A_y)$ , we choose open envelopes for the sets indexed by  $\omega_2$  random test points. Countable support, the generalized  $\Delta$ -system lemma under  $\text{CH}$ , and the fact that there are only  $\aleph_1$  Borel reading types yield  $\omega_2$  independent copies of a single Borel profile. Fresh countable rows of these profiles realize the positive-measure choices produced by the selection lemma. Together with Hechler's counterexample under  $\text{CH}$ , this gives the relative independence of  $P$  from  $\text{ZFC}$  without a large-cardinal hypothesis.

## 1. INTRODUCTION

Let  $\lambda$  and  $\lambda^*$  denote Lebesgue measure and Lebesgue outer measure on  $\mathbb{R}$ . Erdős problem 501 asks whether every family  $(A_y)_{y \in \mathbb{R}}$  of bounded subsets of  $\mathbb{R}$  satisfying

$$\lambda^*(A_y) < 1 \quad (y \in \mathbb{R})$$

has an infinite *independent set*: an infinite  $X \subseteq \mathbb{R}$  such that

$$x \notin A_y$$

for all distinct  $x, y \in X$ . We write  $P$  for the affirmative answer. Erdős and Hajnal proved the existence of arbitrarily large finite independent sets [2]; Hechler proved  $\neg P$  from  $\text{CH}$  [3]; and Newelski, Pawlikowski, and Srećnyński proved the positive answer when the sets  $A_y$  are closed [7]. Lee recently proved the stronger positive conclusion from a countably additive extension of Lebesgue measure to all subsets of  $\mathbb{R}$  [6].

Our main theorem removes the measure-extension and large-cardinal hypotheses.

**Theorem 1.1.** *Let  $M \models \text{ZFC} + \text{CH}$ , let  $\kappa = (\omega_2)^M$ , and let  $G$  be generic over  $M$  for the measure algebra adding  $\kappa$  random reals. In  $M[G]$ , every family  $(A_y)_{y \in \mathbb{R}}$  satisfying*

$$\lambda^*(A_y) < 1 \quad (y \in \mathbb{R})$$

*has an infinite independent set.*

Thus the conclusion is stronger than  $P$ : no boundedness assumption is made. Combining the theorem with Hechler's  $\text{CH}$  counterexample gives the following.

**Corollary 1.2.** *If  $\text{ZFC}$  is consistent, then both  $\text{ZFC} + P$  and  $\text{ZFC} + \neg P$  are consistent.*

The obstacle is that the relation

$$x \in A_y$$

can be completely nonmeasurable. The proof never applies Fubini or Tonelli to this relation. Instead, for many random test points  $y$  we choose open sets  $U_y \supseteq A_y$  of measure below 1. Names for the open sets have countable support. Under  $\text{CH}$ ,  $\omega_2$  such supports can be thinned to a  $\Delta$ -system and then to a common Borel reading type. This produces a measurable directed graph on a profile space. A Tonelli argument supplies a positive-measure set of legal next profiles, and a fresh row of random coordinates hits that set.

This is the random-forcing analogue of the valid part of Lee's argument. Lee obtains a measurable section calculus from a measure defined on every subset of  $\mathbb{R}$ ; here measurability is obtained only after the names have been homogenized. The countable-support and factorization facts used below are the same basic random-forcing tools isolated by Laczkovich and Miller [5]; see also Kunen [4].

For readability we argue with a transitive ground model and a generic point. The proof is a forcing proof and therefore gives the asserted forcing theorem over every model of  $\text{ZFC} + \text{CH}$ .

## 2. A MEASURABLE DIRECTED-GRAPH LEMMA

Let  $(S, \Sigma, \mu)$  be a  $\sigma$ -finite measure space. For a measurable  $E \subseteq S^2$  and  $t, s \in S$ , write

$$E_t = \{s \in S : (t, s) \in E\}, \quad E^s = \{t \in S : (t, s) \in E\}.$$

The direction is chosen so that  $tEs$  means that the point represented by  $t$  is forbidden by the envelope represented by  $s$ .

**Lemma 2.1** (Positive-measure selection). *Suppose  $\mu(S) = \infty$ ,  $E \subseteq S^2$  is measurable, and*

$$(2.1) \quad \mu(E^s) \leq 1 \quad (s \in S).$$

*If  $C \subseteq S$  is measurable and  $\mu(C) = \infty$ , then*

$$(2.2) \quad Q(C) = \{t \in C : \mu(C \setminus E_t) = \infty\}$$

*is measurable and has positive measure.*

*Proof.* The map

$$t \mapsto \mu(C \setminus E_t) = \int_S 1_C(s) 1_{S^2 \setminus E}(t, s) d\mu(s)$$

is measurable by Tonelli, so  $Q(C)$  is measurable.

Suppose toward a contradiction that  $\mu(Q(C)) = 0$ . Since  $\mu(C \setminus Q(C)) = \infty$  and  $\mu$  is  $\sigma$ -finite, choose a measurable  $D \subseteq C \setminus Q(C)$  such that

$$1 < \mu(D) < \infty.$$

For  $k \in \mathbb{N}$ , put

$$D_k = \{t \in D : \mu(C \setminus E_t) \leq k\}.$$

The sets  $D_k$  increase to  $D$ , so for some  $k$ ,

$$d := \mu(D_k) > 1.$$

Choose an increasing measurable exhaustion  $C_0 \subseteq C_1 \subseteq \dots$  of  $C$  by finite-measure sets with  $\mu(C_n) \rightarrow \infty$ . For some  $n$ , writing  $M_n = \mu(C_n)$ , we have

$$(2.3) \quad (M_n - k)d > M_n.$$

For  $t \in D_k$ ,

$$\mu(E_t \cap C_n) = M_n - \mu(C_n \setminus E_t) \geq M_n - k.$$

Tonelli, applied to  $E \cap (D_k \times C_n)$ , gives

$$\begin{aligned} (M_n - k)d &\leq \int_{D_k} \mu(E_t \cap C_n) d\mu(t) \\ &= \int_{C_n} \mu(E^s \cap D_k) d\mu(s) \\ &\leq \int_{C_n} 1 d\mu(s) = M_n, \end{aligned}$$

contradicting (2.3). □

We shall also remove profiles representing the same real. The following one-step formulation records exactly what is needed later.

**Lemma 2.2** (Preservation step). *In addition to the hypotheses of [theorem 2.1](#), let  $x : S \rightarrow \mathbb{R}$  be measurable and suppose every fiber is null:*

$$(2.4) \quad \mu(\{t : x(t) = a\}) = 0 \quad (a \in \mathbb{R}).$$

*If  $C \subseteq S$  is measurable of infinite measure and  $t \in Q(C)$ , then*

$$(2.5) \quad C' = C \setminus (E_t \cup E^t \cup \{s : x(s) = x(t)\})$$

*is measurable and has infinite measure.*

*Proof.* By  $t \in Q(C)$ , the set  $C \setminus E_t$  has infinite measure. The set  $E^t$  has measure at most 1 by [\(2.1\)](#), and the last set in [\(2.5\)](#) is null by [\(2.4\)](#). Removing their union therefore leaves infinite measure.  $\square$

### 3. RANDOM FORCING AND HOMOGENEOUS BOREL READINGS

For a set of coordinates  $\Theta$ , let  $\mathbb{B}(\Theta)$  be the measure algebra of the completion of the product sigma-algebra on  $2^\Theta$ . Every member of the product sigma-algebra depends on only countably many coordinates, so every element of  $\mathbb{B}(\Theta)$  has, modulo null sets, a countably supported representative. We identify a condition with such a positive representative whenever convenient. We say that  $S \subseteq \Theta$  supports a condition or a name if it can be represented using only coordinates from  $S$ . Borel codes introduced below are interpreted canonically in all later intermediate forcing extensions.

#### 3.1. Borel reading of names.

**Lemma 3.1** (Borel reading). *Let  $X$  be a standard Borel space and let  $\dot{z}$  be a  $\mathbb{B}(\Theta)$ -name for an element of  $X$ . There are a countable  $S \subseteq \Theta$  and a Borel map*

$$F : 2^S \rightarrow X$$

*such that*

$$\Vdash_{\mathbb{B}(\Theta)} \dot{z} = F(\dot{G} \restriction S).$$

*Proof.* It is enough to treat  $X = 2^\omega$ . For each  $n$ , choose a Borel representative of the Boolean value  $\|\dot{z}(n) = 1\|$ . That representative has countable coordinate support. The union of these supports over  $n < \omega$  is countable; call it  $S$ . Reading the representatives bit by bit defines the required Borel map. For the general case, code  $X$  as a Borel subset of  $2^\omega$  and modify the reading on the null set where the bitwise value falls outside that subset.  $\square$

For disjoint coordinate sets, random forcing factors in the expected way.

**Lemma 3.2** (Factorization). *Let  $\Sigma$  and  $\Gamma$  be disjoint. If  $G$  is  $\mathbb{B}(\Sigma \cup \Gamma)$ -generic over  $M$ , then  $G \restriction \Sigma$  is  $\mathbb{B}(\Sigma)$ -generic over  $M$  and  $G \restriction \Gamma$  is  $\mathbb{B}(\Gamma)$ -generic over  $M[G \restriction \Sigma]$ . The corresponding statement holds for any finite or countable family of pairwise disjoint coordinate sets.*

*Proof.* This is the usual product decomposition of product measure and its complete measure algebra. Equivalently, it is the random-forcing Fubini theorem; see [\[5, Fact 1 in the proof of Lemma 8\]](#).  $\square$

### 3.2. A generalized $\Delta$ -system lemma.

**Lemma 3.3.** *Assume CH. Every family of  $\omega_2$  countable sets has a  $\Delta$ -subsystem of size  $\omega_2$ .*

*Proof.* Let  $\langle S_\alpha : \alpha < \omega_2 \rangle$  enumerate the family. Take a continuous increasing chain  $\langle M_\xi : \xi < \omega_2 \rangle$  of elementary submodels of a sufficiently large  $H(\theta)$  such that  $|M_\xi| = \omega_1$ ,  $\omega_1 \subseteq M_\xi$ , and the sequence  $\langle S_\alpha : \alpha < \omega_2 \rangle$  belongs to  $M_0$ . Arrange that  $M_{\xi+1} \cap \omega_2$  properly extends  $M_\xi \cap \omega_2$ .

For each  $\xi \in S_{\omega_1}^{\omega_2}$  choose  $\alpha_\xi \in (M_{\xi+1} \cap \omega_2) \setminus M_\xi$ , and put  $A_\xi = S_{\alpha_\xi}$ . Then  $A_\xi \in M_{\xi+1}$ , and since  $A_\xi$  is countable and  $\omega \subseteq M_{\xi+1}$ , we have  $A_\xi \subseteq M_{\xi+1}$ . Let

$$R_\xi = A_\xi \cap M_\xi.$$

Because  $\text{cf}(\xi) = \omega_1$  and  $R_\xi$  is countable, there is  $\eta(\xi) < \xi$  with  $R_\xi \subseteq M_{\eta(\xi)}$ . Fodor's lemma gives a stationary set on which  $\eta(\xi)$  is constant, say equal to  $\eta_*$ . Under CH,

$$|[M_{\eta_*}]^{\aleph_0}| = (\aleph_1)^{\aleph_0} = \aleph_1.$$

Hence one countable set  $R$  is equal to  $R_\xi$  for  $\omega_2$  many  $\xi$ . For two such indices  $\xi < \zeta$ , we have  $A_\xi \subseteq M_\zeta$ , and thus

$$A_\xi \cap A_\zeta = A_\xi \cap (A_\zeta \cap M_\zeta) = A_\xi \cap R = R.$$

These sets form the required  $\Delta$ -system.  $\square$

**3.3. Homogenizing the readings.** We shall use a distinguished countable block inside every support. The next proposition packages all thinning required later.

**Proposition 3.4** (Homogeneous reading). *Assume  $M \models \text{CH}$ , let  $\kappa = (\omega_2)^M$ , and let  $\Theta = \kappa \times \omega$ . For  $\alpha < \kappa$ , put  $D_\alpha = \{\alpha\} \times \omega$ . Let  $p \in \mathbb{B}(\Theta)$ , let  $X$  be a standard Borel space in  $M$ , and for each  $\alpha < \kappa$  let  $\dot{w}_\alpha$  be a name for an element of  $X$ .*

*Then there are a set  $J \subseteq \kappa$  of size  $\kappa$ , a countable root  $R \subseteq \Theta$  supporting  $p$ , pairwise disjoint countable petals  $P_\alpha$  for  $\alpha \in J$ , a fixed countable set  $P$  with a distinguished subset  $D \subseteq P$ , bijections*

$$\pi_\alpha : P \longrightarrow P_\alpha$$

*with  $\pi_\alpha[D] = D_\alpha$ , and a single Borel map*

$$(3.1) \quad F : 2^R \times 2^P \longrightarrow X$$

*such that*

$$(3.2) \quad \Vdash_{\mathbb{B}(\Theta)} \dot{w}_\alpha = F(\dot{G} \restriction R, \pi_\alpha^{-1}(\dot{G} \restriction P_\alpha)) \quad (\alpha \in J).$$

*Moreover,  $D_\alpha \subseteq P_\alpha$  for every  $\alpha \in J$ .*

*Proof.* Choose a countable support  $R_0$  for  $p$ . By [theorem 3.1](#), for each  $\alpha$  choose a countable support  $S_\alpha$  for  $\dot{w}_\alpha$  and enlarge it so that  $R_0 \cup D_\alpha \subseteq S_\alpha$ .

By [theorem 3.3](#), after thinning to a set of indices of size  $\kappa$ , the supports form a  $\Delta$ -system with countable root  $R$ . Since  $R_0$  belongs to every support,  $R_0 \subseteq R$ , so  $R$  supports  $p$ . Put  $P_\alpha = S_\alpha \setminus R$ . The petals are pairwise disjoint. The root meets  $D_\alpha$  for only countably many  $\alpha$ ; discard those indices. Then  $D_\alpha \subseteq P_\alpha$ .

Thin once more so that the cardinality of  $P_\alpha \setminus D_\alpha$  is constant. Choose a fixed countable pair  $(P, D)$  of that type and bijections  $\pi_\alpha : P \rightarrow P_\alpha$  carrying  $D$  onto  $D_\alpha$  and preserving the natural enumeration of the distinguished blocks.

Let  $F_\alpha : 2^R \times 2^{P_\alpha} \rightarrow X$  be a Borel reading of  $\dot{w}_\alpha$ . Pull it back along  $\text{id}_R \cup \pi_\alpha$ . There are only  $2^{\aleph_0} = \aleph_1$  Borel maps from the fixed standard Borel space  $2^R \times 2^P$  to  $X$ . Since there are  $\aleph_2$  indices, a single pulled-back Borel map occurs on a set  $J$  of size  $\aleph_2$ . This map is  $F$ .  $\square$

## 4. OPEN-ENVELOPE PROFILES

Fix  $M \models \text{ZFC} + \text{CH}$ , put  $\kappa = (\omega_2)^M$  and  $\Theta = \kappa \times \omega$ , and let  $\mathbb{B} = \mathbb{B}(\Theta)$ . Suppose toward a contradiction that a condition  $p \in \mathbb{B}$  forces that

$$\dot{\mathcal{A}} = \langle \dot{A}_y : y \in \mathbb{R} \rangle$$

is a counterexample to the conclusion of [theorem 1.1](#).

Fix in  $M$  a Borel measure-preserving map

$$\rho : 2^\omega \longrightarrow [0, 1)$$

whose point fibers are null. For  $\alpha < \kappa$ , let  $\dot{r}_\alpha$  be the random real read from  $D_\alpha = \{\alpha\} \times \omega$ , and for  $m \in \mathbb{Z}$  define

$$(4.1) \quad \dot{x}_{\alpha,m} = m + \rho(\dot{r}_\alpha).$$

Thus, for fixed  $m$ , these names are uniformly distributed on  $I_m = [m, m+1)$ .

Let  $\mathcal{O}$  be a fixed standard Borel coding space for open subsets of  $\mathbb{R}$ . We write  $U(c)$  for the open set coded by  $c \in \mathcal{O}$ . The relation  $x \in U(c)$  is Borel in  $(x, c)$ , and  $c \mapsto \lambda(U(c))$  is Borel.

For each  $\alpha < \kappa$  and  $m \in \mathbb{Z}$ , outer regularity and the forcing maximum principle give an open-code name  $\dot{c}_{\alpha,m}$  such that

$$(4.2) \quad p \Vdash \dot{A}_{\dot{x}_{\alpha,m}} \subseteq U(\dot{c}_{\alpha,m}) \quad \text{and} \quad \lambda(U(\dot{c}_{\alpha,m})) < 1.$$

Mixing with a fixed default code off  $p$ , we may and do arrange that the top condition forces  $\dot{c}_{\alpha,m} \in \mathcal{O}$ . Bundle the countable sequence into

$$\dot{w}_\alpha = \langle \dot{c}_{\alpha,m} : m \in \mathbb{Z} \rangle \in \mathcal{O}^\mathbb{Z}.$$

Apply [theorem 3.4](#). We obtain  $J, R, P, D$ , the petals  $P_\alpha$ , the identifications  $\pi_\alpha$ , and a common Borel map

$$F = \langle F_m : m \in \mathbb{Z} \rangle : 2^R \times 2^P \longrightarrow \mathcal{O}^\mathbb{Z}.$$

Let  $G$  be an  $M$ -generic point containing  $p$ , put

$$g = G \restriction R, \quad N = M[g],$$

and for  $\alpha \in J$  define the normalized petal profile

$$(4.3) \quad z_\alpha = \pi_\alpha^{-1}(G \restriction P_\alpha) \in \Omega := 2^P.$$

By [theorem 3.2](#), the family  $(z_\alpha)_{\alpha \in J}$  consists of mutually random points of  $(\Omega, \nu)$  over  $N$ , where  $\nu$  is product measure.

For  $z \in \Omega$  and  $m \in \mathbb{Z}$ , let

$$(4.4) \quad U_m(z) = U(F_m(g, z)).$$

The next lemma is the conditional-measure step that converts the forced measure bound into a conull collection of profiles.

**Lemma 4.1.** *In  $N$ , the set*

$$(4.5) \quad \Omega_0 = \{z \in \Omega : \lambda(U_m(z)) < 1 \text{ for every } m \in \mathbb{Z}\}$$

*is Borel and conull.*

*Proof.* For fixed  $m$ , let

$$B_m = \{(r, z) \in 2^R \times 2^P : \lambda(U(F_m(r, z))) \geq 1\}.$$

This is Borel. By (4.2) and (3.2), after identifying  $p$  with its Borel representative in  $2^R$ , the set

$$B_m \cap (p \times 2^P)$$

has product measure zero. Fubini implies that

$$N_m = \{r \in p : \nu((B_m)_r) > 0\}$$

is a null Borel subset of  $2^R$ . The random generic  $g$  avoids  $\bigcup_{m \in \mathbb{Z}} N_m$ . Hence for every  $m$  the bad  $z$ -section at  $g$  is null, and their countable union is null. This is exactly (4.5).  $\square$

For every actual profile  $z_\alpha$  and every  $m \in \mathbb{Z}$ , the forcing theorem, (4.2), and the common reading give

$$(4.6) \quad A_{m+\rho(z_\alpha \upharpoonright D)} \subseteq U_m(z_\alpha).$$

This assertion is required only for the actual generic profiles. We make no claim that  $U_m(z)$  contains the corresponding  $A_y$  for an arbitrary  $z \in \Omega$ .

## 5. THE PROFILE GRAPH AND THE FRESH-ROW RECURSION

Work first in the root extension  $N$ . Put

$$S = \mathbb{Z} \times \Omega_0, \quad \mu = \text{counting measure} \times \nu.$$

Then  $\mu(S) = \infty$ . For  $t = (m, z) \in S$ , define

$$(5.1) \quad x(t) = m + \rho(z \upharpoonright D), \quad U(t) = U_m(z).$$

Define a Borel directed graph  $E \subseteq S^2$  by

$$(5.2) \quad (t, s) \in E \iff x(t) \in U(s).$$

**Lemma 5.1.** *For every  $s \in S$ ,*

$$(5.3) \quad \mu(E^s) = \lambda(U(s)) < 1.$$

*Moreover, every fiber of  $x : S \rightarrow \mathbb{R}$  is null.*

*Proof.* Let  $s = (n, w)$ . Since  $z \mapsto m + \rho(z \upharpoonright D)$  has Lebesgue distribution on  $I_m = [m, m+1)$  and  $\Omega_0$  is conull,

$$\begin{aligned} \mu(E^s) &= \sum_{m \in \mathbb{Z}} \nu\{z \in \Omega_0 : m + \rho(z \upharpoonright D) \in U_n(w)\} \\ &= \sum_{m \in \mathbb{Z}} \lambda(U_n(w) \cap I_m) \\ &= \lambda(U_n(w)) < 1. \end{aligned}$$

A real  $a$  belongs to a unique interval  $I_m$ , and the fiber of  $a - m$  under  $\rho$  is null. Therefore  $\mu(\{t : x(t) = a\}) = 0$ .  $\square$

Choose in  $M$  pairwise distinct indices

$$\alpha_{j,k} \in J \quad (j, k < \omega)$$

and write  $z_{j,k} = z_{\alpha_{j,k}}$ . For  $j < \omega$ , let

$$(5.4) \quad N_j = N[\langle z_{i,k} : i < j, k < \omega \rangle].$$

By factorization, the row

$$\vec{z}_j = \langle z_{j,k} : k < \omega \rangle$$

is  $\nu^\omega$ -random over  $N_j$ . Throughout the recursion, the Borel codes for  $S$ ,  $E$ ,  $\Omega_0$ , and earlier candidate sets are interpreted in the current intermediate model. The defining identities in [theorem 5.1](#) are absolute for these codes, so the section bound and the null-fiber assertion remain valid for newly added profiles.

We recursively construct Borel sets  $C_j \subseteq S$  and actual profiles  $t_j \in C_j$ . Start with  $C_0 = S$ . Suppose  $C_j \in N_j$  is Borel and  $\mu(C_j) = \infty$ . By [theorem 2.1](#),

$$Q_j = Q(C_j) = \{t \in C_j : \mu(C_j \setminus E_t) = \infty\}$$

is Borel and has positive measure. Hence for some  $m_j \in \mathbb{Z}$  the Borel set

$$(5.5) \quad H_j = \{z \in \Omega_0 : (m_j, z) \in Q_j\}$$

has positive  $\nu$ -measure. Choose the first such  $m_j$  in a fixed enumeration of  $\mathbb{Z}$ .

The event that every coordinate of a  $\nu^\omega$ -random sequence avoids  $H_j$  has measure

$$\lim_{n \rightarrow \infty} (1 - \nu(H_j))^n = 0.$$

Since  $\tilde{z}_j$  is random over  $N_j$ , there is a least  $k_j$  such that  $z_{j,k_j} \in H_j$ . Set

$$(5.6) \quad t_j = (m_j, z_{j,k_j})$$

and define

$$(5.7) \quad C_{j+1} = C_j \setminus (E_{t_j} \cup E^{t_j} \cup \{s : x(s) = x(t_j)\}).$$

By [theorem 2.2](#),  $C_{j+1}$  has infinite measure. It is Borel and belongs to  $N_{j+1}$ , so the recursion continues.

Put

$$y_j = x(t_j) = m_j + \rho(z_{j,k_j} \upharpoonright D).$$

The fiber removal in (5.7) makes the  $y_j$  pairwise distinct. Let  $i < j$ . Since  $t_j \in C_{i+1}$ ,

$$t_j \notin E^{t_i} \quad \text{and} \quad t_j \notin E_{t_i}.$$

The first relation says

$$(t_j, t_i) \notin E, \quad \text{hence} \quad y_j \notin U(t_i).$$

The second says

$$(t_i, t_j) \notin E, \quad \text{hence} \quad y_i \notin U(t_j).$$

Each  $t_i$  is the actual profile associated with  $\alpha_{i,k_i}$ . Therefore (4.6) yields

$$A_{y_i} \subseteq U(t_i), \quad A_{y_j} \subseteq U(t_j).$$

Consequently

$$y_j \notin A_{y_i} \quad \text{and} \quad y_i \notin A_{y_j}.$$

Thus  $\{y_j : j < \omega\}$  is an infinite independent set, contradicting the choice of  $p$ . This proves [theorem 1.1](#).

## 6. CONSEQUENCES AND METHODOLOGICAL REMARKS

*Proof of theorem 1.2.* If ZFC is consistent, take a model of ZFC and pass to its constructible universe, which satisfies CH. Forcing with the measure algebra adding  $\omega_2$  random reals gives a model of  $P$  by [theorem 1.1](#). On the other hand, CH implies  $\neg P$  by Hechler's theorem. Hence both sides are consistent relative to ZFC.  $\square$

For completeness, the elementary CH counterexample can be written as follows. Enumerate  $\mathbb{R} = \{r_\alpha : \alpha < \omega_1\}$ , and for  $y = r_\beta$  put

$$A_y = \{r_\alpha : \alpha < \beta \text{ and } |r_\alpha| \leq |y| + 1\}.$$

Each  $A_y$  is countable and bounded. If  $x_0 \prec x_1 \prec \dots$  were an increasing sequence from an infinite independent set, then

$$|x_i| > |x_j| + 1 \quad (i < j),$$

which is impossible.

The proof of [theorem 1.1](#) uses no regularity of the original sets

$$B_x = \{y : x \in A_y\}.$$

In particular, it does not seek an inner-measure-large complement of  $B_x$ . All applications of Tonelli concern the Borel relation [\(5.2\)](#) after the envelope names have been homogenized. The arbitrary family enters again only through the separately forced containments [\(4.6\)](#) at the countably many generic profiles selected by the recursion.

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